

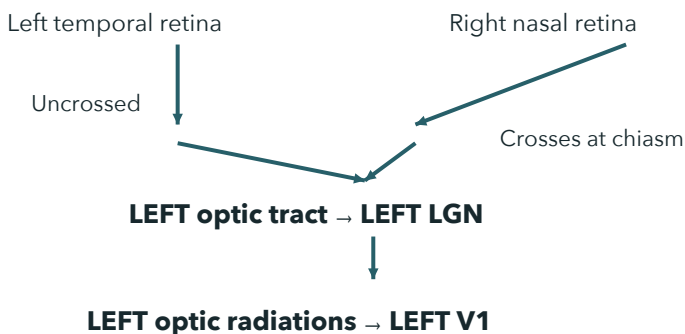
Visual pathways and pupil reflexes

Use the field pattern and both pupils to localise the lesion

Visual field first, anatomical side second

One hemifield reaches the opposite hemisphere

RIGHT visual field



Nasal retinal fibres cross at the optic chiasm. Temporal retinal fibres remain ipsilateral. Each optic tract carries the opposite visual hemifield from both eyes.

Keep field and retina distinct

- Temporal visual field projects to nasal retina; nasal visual field projects to temporal retina. The retinal image is also inverted vertically.
- The right visual hemifield reaches the left optic tract and left visual cortex. The left hemifield reaches the right side.
- Do not equate left-eye blindness with loss of the left visual hemifield. One is monocular; the other can involve both eyes.

The conscious visual route

- Photoreceptors → bipolar cells → retinal ganglion cells
- Ganglion-cell axons form optic nerve → chiasm → optic tract
- Synapse in lateral geniculate nucleus of the thalamus
- Optic radiations → primary visual cortex beside calcarine sulcus

Cortical map

Superior visual field projects below the calcarine sulcus; inferior field projects above it. Central vision occupies a disproportionately large, posterior cortical area near the occipital pole. Extrastriate dorsal processing supports spatial position and motion; ventral processing supports object and face recognition. For lesion questions, trace the missing visual field before choosing the side of the brain.

Localisation from visual field patterns

Pattern and pathway segment

Lesion	Expected field defect
Left optic nerve	Left monocular visual loss, partial or complete
Midline optic chiasm	Bitemporal hemianopia from crossing nasal fibres
Left optic tract	Right homonymous hemianopia
Left temporal radiation, Meyer's loop	Right superior homonymous quadrantanopia
Left parietal radiation	Right inferior homonymous quadrantanopia
Left primary visual cortex	Right homonymous defect; central field may be spared

What the pattern tells you

- Monocular loss usually localises anterior to the chiasm. A homonymous defect localises behind it, on the side opposite the missing visual space.
- Bitemporal loss suggests a chiasmal lesion; a pituitary mass can compress the chiasm from below.
- Temporal radiation carries superior field information. Parietal radiation carries inferior field information.
- More posterior lesions often produce more congruous defects, meaning a closer match between the two eyes.

Macular sparing

An occipital lesion may spare central vision when the occipital pole remains functional, often through preserved collateral blood supply. This supports an occipital localisation in the appropriate setting but is not an absolute rule; cortical lesions can also split the central field.

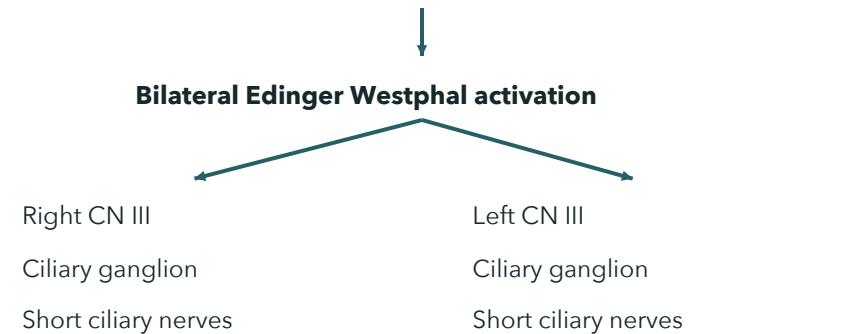
Check the description before answering

“Left homonymous hemianopia” means loss of the left half of visual space in each eye. It does not mean both nasal fields or blindness confined to the left eye. A quadrant defect preserves the other quadrant on the affected side.

Light reflex one input activates two outputs

Bilateral output explains the consensual reflex

Retina → optic nerve and tract → pretectal area



Both sphincter pupillae constrict

Retinal input reaches the pretectal region. Each pretectal nucleus drives both Edinger-Westphal nuclei; each pupil then uses its own CN III efferent pathway.

Each pupil has its own efferent arm

- CN III carries preganglionic parasympathetic fibres to a synapse in the ciliary ganglion.
- Postganglionic short ciliary nerves activate sphincter pupillae. The reflex branches from the optic tract before the LGN.

Observe each pupil separately

Term	What is observed
Direct response	Constriction in the illuminated eye
Consensual response	Constriction in the fellow eye
Afferent integrity	Does illuminating this eye drive both pupils?
Efferent integrity	Can this pupil constrict, whichever eye is illuminated?

Conscious sight is a separate destination

The light reflex branches from the pathway before the LGN and does not require the visual cortex. Bilateral occipital lesions can therefore abolish conscious vision while preserving pupillary light responses. The near response uses cortical connections and couples accommodation, convergence and miosis. A pupil that constricts better to near effort than to light shows light-near dissociation; this finding requires localisation using the rest of the examination.

Separate weak input from a failed pupil output

Idealised unilateral lesion patterns

Lesion	Light in right eye	Light in left eye
Right optic nerve afferent damage	Both constrict weakly	Both constrict well
Right parasympathetic efferent failure	Right stays large; left constricts	Right stays large; left constricts

RAPD compare the afferent inputs

- Swing the light between eyes using comparable illumination. Keep fixation distant to reduce a near response.
- When moving from the better eye to an eye with a relative afferent defect, both pupils constrict less or appear to dilate.
- An RAPD usually indicates asymmetric optic nerve disease or extensive asymmetric retinal disease. It need not produce anisocoria.
- A negative RAPD does not exclude equal bilateral afferent damage. If one pupil cannot move, observe the other pupil while alternating the light.

Anisocoria identify the abnormal pupil

- Compare pupil sizes in bright and dim lighting
- Difference greater in bright light → larger pupil fails to constrict → consider parasympathetic or iris dysfunction
- Difference greater in darkness → smaller pupil fails to dilate → consider sympathetic dysfunction, such as Horner syndrome

Add the associated signs

Horner syndrome pairs miosis with mild ptosis and sometimes anhidrosis; the light response is usually preserved. A pupil-involving CN III palsy may pair mydriasis with marked ptosis and a down-and-out eye. New painful Horner syndrome or a new pupil-involving CN III palsy warrants urgent assessment because vascular compression or carotid dissection may be responsible.

Field atlas and pupil test

A Read the lost field

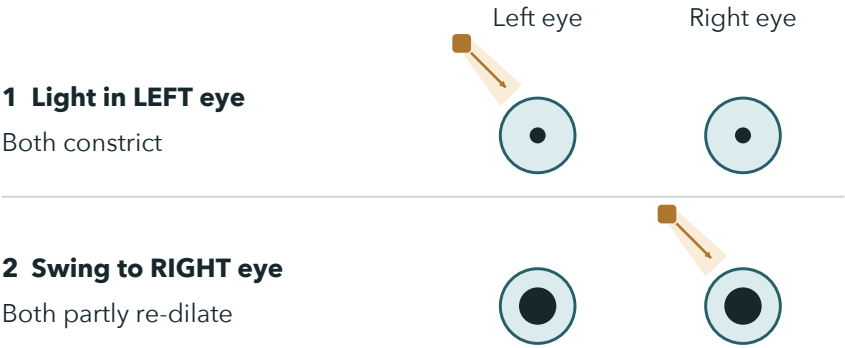
Lesion site	Left eye	Right eye
Left optic nerve Left-eye loss		
Midline chiasm Bitemporal loss		
Left tract / cortex Right hemifield loss		
Left temporal radiation Right upper quadrant		
Left parietal radiation Right lower quadrant		

■ Shaded = lost field; view as seen by patient

Each circle is a visual field, not a retina. Left-sided retrochiasmal lesions affect right visual space in both eyes.

B Compare the two afferent inputs

Example: RIGHT RAPD; intact pupil outputs



Right RAPD: moving equal illumination from the better left eye to the weaker right eye reduces drive to both pupils. Anisocoria is not required.

Reference map

A: TBL 2 slides 17-25; NIH visual pathway and Meyer loop study. B: slides 8-11; AAO RAPD and NIH light-reflex pathway. Original schematics.